

Q1.

Over a period of time it has been shown that the mean number of vehicles passing a service station on a motorway is 50 per minute.

After a new motorway junction was built nearby, Xander observed that 30 vehicles passed the service station in one minute.

- (a) Xander claims that the construction of the new motorway junction has reduced the mean number of vehicles passing the service station per minute.

Investigate Xander's claim, using a suitable test at the 1% level of significance.

(6)

- (b) For your test carried out in part (a) state, in context, the meaning of a Type 1 error.

(1)

- (c) Explain why the model used in part (a) might be invalid.

(1)

(Total 8 marks)

Q2.

The number of printers, V , bought during one day from the *Verigood* store can be modelled by a Poisson distribution with mean 4.5

The number of printers, W , bought during one day from the *Winnerprint* store can be modelled by a Poisson distribution with mean 5.5

- (a) Find the probability that the total number of printers bought during one day from *Verigood* and *Winnerprint* stores is greater than 10.

(2)

- (b) State the circumstance under which the distributional model you used in part (a) would not be valid.

(1)

(Total 3 marks)

Q3.

A large office block is busy during the five weekdays, Monday to Friday, and less busy during the two weekend days, Saturday and Sunday. The block is illuminated by fluorescent light tubes which frequently fail and must be replaced with new tubes by John, the caretaker.

The number of fluorescent tubes that fail on a particular weekday can be modelled by a Poisson distribution with mean 1.5.

The number of fluorescent tubes that fail on a particular weekend day can be modelled by a Poisson distribution with mean 0.5.

- (a) Find the probability that:

- (i) on one particular Monday, exactly 3 fluorescent light tubes fail;

(2)

- (ii) during the two days of a weekend, more than 1 fluorescent light tube fails; (2)
 - (iii) during a complete seven-day week, fewer than 10 fluorescent light tubes fail. (4)
- (b) John keeps a supply of new fluorescent light tubes. More new tubes are delivered every Monday morning to replace those that he has used during the previous week. John wants the probability that he runs out of new tubes before the next Monday morning to be less than 1 per cent. Find the minimum number of new tubes that he should have available on a Monday morning. (2)
- (c) Give a reason why a Poisson distribution with mean 0.375 is unlikely to provide a satisfactory model for the number of fluorescent light tubes that fail between 1 am and 7 am on a weekday. (1)
- (Total 11 marks)**

Q4.

Gamma-ray bursts (GRBs) are pulses of gamma rays lasting a few seconds, which are produced by explosions in distant galaxies. They are detected by satellites in orbit around Earth. One particular satellite detects GRBs at a constant average rate of 3.5 per week (7 days).

You may assume that the detection of GRBs by this satellite may be modelled by a Poisson distribution.

- (a) Find the probability that the satellite detects:
- (i) exactly 4 GRBs during one particular week; (2)
 - (ii) at least 2 GRBs on one particular day; (3)
 - (iii) more than 10 GRBs but fewer than 20 GRBs during the 28 days of February 2013. (3)
- (b) Give **one** reason, apart from the constant average rate, why it is likely that the detection of GRBs by this satellite may be modelled by a Poisson distribution. (1)
- (Total 9 marks)**

Q5.

Lucy is the captain of her school's cricket team.

The number of catches, X , taken by Lucy during any particular cricket match may be modelled by a Poisson distribution with mean 0.6.

The number of run-outs, Y , effected by Lucy during any particular cricket match may be modelled by a Poisson distribution with mean 0.15.

- (a) Find:
- (i) $P(X \leq 1)$; (1)
- (ii) $P(X \leq 1 \text{ and } Y \geq 1)$. (4)
- (b) State the assumption that you made in answering part (a)(ii). (1)
- (c) During a particular season, Lucy plays in 16 cricket matches.
- (i) Calculate the probability that the number of catches taken by Lucy during this season is exactly 10. (2)
- (ii) Determine the probability that the total number of catches taken and run-outs effected by Lucy during this season is at least 15. (3)
- (Total 11 marks)

Q6.

The number of cars passing a speed camera on a main road between 9.30 am and 11.30 am may be modelled by a Poisson distribution with a mean rate of 2.6 per minute.

- (a) (i) Write down the distribution of X , the number of cars passing the speed camera during a 5-minute interval between 9.30 am and 11.30 am. (1)
- (ii) Determine $P(X = 20)$. (2)
- (iii) Determine $P(6 \leq X \leq 18)$. (3)
- (b) Give **two** reasons why a Poisson distribution with mean 2.6 may not be a suitable model for the number of cars passing the speed camera during a 1-minute interval between 8.00 am and 9.30 am on weekdays. (2)
- (c) When n cars pass the speed camera, the number of cars, Y , that exceed 60 mph may be modelled by the distribution $B(n, 0.2)$.
- Given that $n = 20$, determine $P(Y \geq 5)$. (2)
- (d) Stating a necessary assumption, calculate the probability that, during a given 5-minute interval between 9.30 am and 11.30 am, exactly 20 cars pass the speed camera of which at least 5 are exceeding 60 mph. (3)
- (Total 13 marks)

Q7.

The random variable Y has a Poisson distribution with $E(Y) = 2.5$.

Given that $Z = 4Y + 30$:

- (a) show that $\text{Var}(Z) = E(Z)$;

(3)

- (b) give a reason why the distribution of Z is **not** Poisson.

(1)

(Total 4 marks)

Q8.

The number of telephone calls received, during an 8-hour period, by an IT company that request an urgent visit by an engineer may be modelled by a Poisson distribution with a mean of 7.

- (a) Determine the probability that, during a given 8-hour period, the number of telephone calls received that request an urgent visit by an engineer is:

- (i) at most 5;

(1)

- (ii) exactly 7;

(2)

- (iii) at least 5 but fewer than 10.

(3)

- (b) Write down the distribution for the number of telephone calls received each hour that request an urgent visit by an engineer.

(1)

- (c) The IT company has 4 engineers available for urgent visits and it may be assumed that each of these engineers takes exactly 1 hour for each such visit.

At 10 am on a particular day, all 4 engineers are available for urgent visits.

- (i) State the maximum possible number of telephone calls received between 10 am and 11 am that request an urgent visit and for which an engineer is immediately available.

(1)

- (ii) Calculate the probability that at 11 am an engineer will **not** be immediately available to make an urgent visit.

(4)

- (d) Give a reason why a Poisson distribution may not be a suitable model for the number of telephone calls per hour received by the IT company that request an urgent visit by an engineer.

(1)

(Total 13 marks)

Q9.

John works from home. The number of business letters, X , that he receives on a weekday may be modelled by a Poisson distribution with mean 5.0.

The number of private letters, Y , that he receives on a weekday may be modelled by a Poisson distribution with mean 1.5.

(a) Find, for a given weekday:

(i) $P(X < 4)$;

(2)

(ii) $P(Y = 4)$.

(2)

(b) (i) Assuming that X and Y are independent random variables, determine the probability that, on a given weekday, John receives a **total** of more than 5 business and private letters.

(3)

(ii) Hence calculate the probability that John receives a **total** of more than 5 business and private letters on at least 7 out of 8 given weekdays.

(3)

(c) The numbers of letters received by John's neighbour, Brenda, on 10 consecutive weekdays are

15 8 14 7 6 8 2 8 9 3

(i) Calculate the mean and the variance of these data.

(2)

(ii) State, giving a reason based on your answers to part (c)(i), whether or not a Poisson distribution might provide a suitable model for the number of letters received by Brenda on a weekday.

(2)

(Total 14 marks)

Q10.

A new information technology centre is advertising places on its one-week residential computer courses.

(a) The number of places, X , booked each week on the publishing course may be modelled by a Poisson distribution with a mean of 9.0.

(i) State the standard deviation of X .

(1)

(ii) Calculate $P(6 < X < 12)$.

(3)

(b) The number of places booked each week on the web design course may be modelled by a Poisson distribution with a mean of 2.5.

- (i) Write down the distribution for T , the **total** number of places booked each week on the publishing and web design courses. (1)
- (ii) Hence calculate the probability that, during a given week, a total of fewer than 2 places are booked. (3)
- (c) The number of places booked on the database course during each of a random sample of 10 weeks is as follows:
- 14 15 8 16 18 4 10 12 15 8
- By calculating appropriate numerical measures, state, with a reason, whether or not the Poisson distribution $Po(12.0)$ could provide a suitable model for the number of places booked each week on the database course. (3)
- (Total 11 marks)**

Q11.

The number of computers, A , bought during one day from the Amplebuy computer store can be modelled by a Poisson distribution with a mean of 3.5.

The number of computers, B , bought during one day from the Bestbuy computer store can be modelled by a Poisson distribution with a mean of 5.0.

- (a) (i) Calculate $P(A = 4)$. (2)
- (ii) Determine $P(B \leq 6)$. (1)
- (iii) Find the probability that a total of fewer than 10 computers is bought from these two stores on one particular day. (3)
- (b) Calculate the probability that a total of fewer than 10 computers is bought from these two stores on at least 4 out of 5 consecutive days. (3)
- (c) The numbers of computers bought from the Choicebuy computer store over a 10-day period are recorded as
- 8 12 6 6 9 15 10 8 6 12
- (i) Calculate the mean and variance of these data. (2)
- (ii) State, giving a reason based on your results in part (c) (i), whether or not a Poisson distribution provides a suitable model for these data. (2)
- (Total 13 marks)**

Q12.

A study undertaken by Goodhealth Hospital found that the number of patients each month, X , contracting a particular superbug can be modelled by a Poisson distribution with a mean of 1.5.

- (a) (i) Calculate $P(X = 2)$. (2)
- (ii) Hence determine the probability that exactly 2 patients will contract this superbug in each of three consecutive months. (2)
- (b) (i) Write down the distribution of Y , the number of patients contracting this superbug in a given 6-month period. (1)
- (ii) Find the probability that at least 12 patients will contract this superbug during a given 6-month period. (2)
- (c) State **two** assumptions implied by the use of a Poisson model for the number of patients contracting this superbug. (2)
- (Total 9 marks)**

Q13.

The number of A-grades, X , achieved in total by students at Lowkey School in their Mathematics examinations each year can be modelled by a Poisson distribution with a mean of 3.

- (a) Determine the probability that, during a 5-year period, students at Lowkey School achieve a total of more than 18 A-grades in their Mathematics examinations. (3)
- (b) The number of A-grades, Y , achieved in total by students at Lowkey School in their English examinations each year can be modelled by a Poisson distribution with a mean of 7.
- (i) Determine the probability that, during a year, students at Lowkey School achieve a total of fewer than 15 A-grades in their Mathematics and English examinations. (3)
- (ii) What assumption did you make in answering part (b)(i)? (1)
- (Total 7 marks)**

Q14.

In the Parkside Sunday Football League, the number of goals scored in each match may be modelled by Poisson distributions.

- (a) When the weather is dry, the mean number of goals scored is 1.6 per 60 minutes of

play.

Find the probability that on a dry Sunday:

- (i) exactly 3 goals are scored during the first 60 minutes of a particular match; (2)
 - (ii) a match lasting **90 minutes** ends with the score 0–0; (2)
 - (iii) when 5 matches, each lasting 90 minutes, are played, a total of 20 or more goals is scored. (3)
- (b) When the weather is wet, the mean number of goals scored is 1.2 per 60 minutes of play.

For a match lasting 90 minutes played on a wet Sunday:

- (i) find the standard deviation of the number of goals scored; (2)
 - (ii) explain why there is insufficient data to calculate the probability of the final score being 3–0. (1)
- (c) A statistician plans to examine the distribution of the number of goals scored in all matches played during the 2004/5 season. Explain why a Poisson distribution is unlikely to provide an adequate model for this distribution. (1)
- (Total 11 marks)**

Q15.

- (a) Suggest a distribution which might be appropriate for modelling the number of new bicycles sold per day by a bicycle shop. (1)
- (b) The shop sells the following numbers of new bicycles on successive days.
- | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 9 | 1 | 6 | 3 | 4 | 6 | 4 | 5 | 1 | 0 | 5 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
- (i) Calculate the mean and the variance of these data. (2)
 - (ii) State, giving a reason, whether your calculations in part (b)(i) support the model you have suggested in part (a). (2)
- (Total 5 marks)**

Q16.

The number of family groups entering a museum on an August afternoon may be modelled by a Poisson distribution with mean 0.60 per minute.

- (a) Find the probability that during a particular minute 2 or fewer such groups enter the

museum.

(2)

- (b) Find the probability that a particular **five-minute** interval more than 4 such groups enter the museum.

(3)

- (c) Explain why a Poisson distribution would not provide a suitable model for the number of **visitors** entering the museum during one-minute intervals on an August afternoon.

(2)

(Total 7 marks)

Q17.

The number of cyclists passing a point on a cycle track alongside a main road into Oxford, during the morning rush hour, may be modelled by a Poisson distribution with mean 2.8 per minute.

- (a) Find the probability that the number of cyclists passing the point during a particular minute in the morning rush hour is:

(i) one or fewer;

(ii) 3 or more.

(3)

- (b) Find the probability that fewer than ten cyclists pass the point during a **five** minute interval in the morning rush hour.

(3)

- (c) The numbers of cyclists passing the point during ten, randomly selected, one minute intervals over a twenty four hour period, were:

0 0 5 1 0 4 1 0 0 2

- (i) Calculate the mean and variance of this sample.

(2)

- (ii) Give a reason, based on your results in part (c)(i), why the Poisson distribution is unlikely to provide an adequate model for the number of cyclists passing the point over a twenty four hour period.

(1)

- (iii) Give a reason, **not** based on your results in part (c)(i), why the Poisson distribution is unlikely to provide an adequate model for the number of cyclists passing the point over a twenty four hour period.

(1)

(Total 10 marks)

Q18.

Bronwen runs a post office in a large village. The number of registered letters posted at this office may be modelled by a Poisson distribution with mean 1.4 per day.

- (a) Find the probability that at this post office:

- (i) two of fewer registered letters are posted on a particular day;
- (ii) a total of four or more registered letters are posted on two consecutive days.

(5)

- (b) The village also contains a post office run by Gopal. Here the number of registered letters posted may be modelled by a Poisson distribution with mean 2.4 per day.

Find the probability that, on a particular day, the number of registered letters posted at Gopal's post office is less than 4.

(2)

- (c) The numbers of registered letters posted at the two post offices are independent. Find the probability that, on a particular day, the total number of registered letters posted at the two post offices is more than six.

(3)

- (d) Give one reason why the Poisson distribution might **not** provide a suitable model for the number of registered letters posted daily at a post office.

(1)

(Total 11 marks)



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